Guanyu Hou

Trustworthy AI

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RESEARCH INTERESTS

My research focuses on Trustworthy AI, with two main directions: (i) **automated red-teaming of large language models and image/video generation models**, including audio injection attacks and temporal-aware red-teaming for text-to-video models; and (ii) **multimodal backdoors, data theft, and privacy protection**, covering multimodal in-context learning, conditional backdoors in large language models, privacy protection in retrieval-augmented generation (RAG), and protection against unauthorized video personalization.

EDUCATION

The University of Manchester — Master
MSc Artificial Intelligence — 2025 – 2026
GPA: 73.75/100

Oxford Brookes University — Bachelor
BSc with First Class Honours in Software Engineering — 2021 – 2025
Class Rank: 4/99 GPA: 76.9/100
Completed through the officially approved 4+0 joint education programme between Oxford Brookes University and Chengdu University of Technology (CDUT)
Also awarded a Bachelor of Engineering in Software Engineering by CDUT.

INTERNSHIP EXPERIENCE

Shanghai Artificial Intelligence Laboratory (PJLab) — Shanghai, China
Research Intern (Advisor: Lijun Li) — Sep 2026 – Present

- Focus on AI4Science safety, investigating vulnerability detection and mechanisms for establishing safety boundaries in scientific domains.

ByteDance — Beijing, China
Algorithm Intern (Intelligent Security) — Jul 2026 – Sep 2026

- **Multilingual Cascaded Safety Classifier:** Optimized the first-stage classifier in a cascaded LLM firewall for global AI products through supervised fine-tuning (SFT), automated data cleaning, and multilingual data augmentation. Achieved **precision, recall, and F1 scores above 97%** in multilingual evaluations and successfully deployed the model to production.
- **Optimization for Long-Reasoning Samples:** Evaluated input preprocessing and training-time data augmentation strategies, improving **both precision and recall by approximately 1 percentage point**.
- **Jailbreak Defense Enhancement:** Analyzed red-team jailbreak test logs to identify the root causes of missed detections. Constructed targeted, challenging adversarial examples through automated prompt optimization and manual verification, expanded the core internal dataset, and validated robustness in controlled experiments, successfully blocking all tested jailbreak inputs of the same types.

PUBLICATIONS

Seven accepted or published papers: two at CCF-A venues, three at CCF-B venues, one at a CCF-C venue, and one in a JCR Q2 journal. These include ***one sole first-author paper and two co-first-author papers****, with two papers selected for oral presentations. One additional manuscript is under review and available as an arXiv preprint. Full-text links are available on my personal website.*

[**EMNLP 2025**] **(CCF-B)** Evaluating Robustness of Large Audio Language Models to Audio Injection: An Empirical Study. *(Accepted Jul 2025)*
Guanyu Hou*, Jiaming He* (** Equal contribution*), Yinhang Zhou, Ji Guo, Yitong Qiao, Rui Zhang, Wenbo Jiang

[**CVPR 2026**] **(CCF-A)** TEAR: Temporal-aware Automated Red-teaming for Text-to-Video Models. *(Accepted Feb 2026)* **(Oral, 0.88%)**
Jiaming He, **Guanyu Hou**, Hongwei Li, Zhicong Huang, Kangjie Chen, Yi Yu, Wenbo Jiang, Guowen Xu, Tianwei Zhang

[**ICASSP 2025**] **(CCF-B)** PRESS: Defending Privacy in Retrieval-Augmented Generation via Embedding Space Shifting. *(Accepted Dec 2024)*
Jiaming He*, Cheng Liu*, **Guanyu Hou*** (** Equal contribution*), Wenbo Jiang, Jiachen Li

[**MM Asia 2025**] **(CCF-C)** When Hallucinated Concepts Cross Modals: Unveiling Backdoor Vulnerability in Multi-modal In-context Learning. *(Published Dec 2025)*
Guanyu Hou, Jiaming He, Yitong Qiao, Jiachen Li, Qiyang Song, Ji Guo, Zihan Wang, Wenbo Jiang

[**AAAI 2025**] **(CCF-A)** Watch Out for Your Guidance on Generation! Exploring Conditional Backdoor Attacks against Large Language Models. *(Accepted Nov 2024)* **(Oral, 4.6%)**
Jiaming He, Wenbo Jiang, **Guanyu Hou**, Wenshu Fan, Rui Zhang, Hongwei Li

[**ICME 2025**] **(CCF-B)** Weaponizing Tokens: Backdooring Text-to-Image Generation via Token Remapping. *(Accepted Mar 2025)*
Jiaming He, Wenbo Jiang, **Guanyu Hou**, Qiyang Song, Ji Guo, Hongwei Li

[**Electronics**] **(JCR Q2)** Data Stealing Attacks against Large Language Models via Backdooring. *(Accepted Jul 2024)*
Jiaming He, **Guanyu Hou**[†], Xinyue Jia, Yangyang Chen, Wenqi Liao, Yinhang Zhou, Rang Zhou[†] ([†] *Corresponding authors*)

[**arXiv 2026**] ChronoLock: Protecting Videos from Unauthorized Text-to-Video Personalization. *(Preprint, arXiv:2606.21146; under review)*
Jiaming He, Jiashu Zhang, **Guanyu Hou**, Shuhan Ye, Hanwei Zhu, Yi Yu, Xudong Jiang

ACADEMIC SERVICE

- Reviewer, AAAI 2026
- Reviewer, NeurIPS 2026
- Program Committee Member, AAAI 2027

LANGUAGE PROFICIENCY

English

IELTS Overall Band Score: 7.0

SKILLS & TOOLS

Machine Learning Frameworks and Tools

Frameworks: PyTorch, scikit-learn
Tools: Hugging Face (Transformers, PEFT, TRL), LLaMA-Factory, vLLM

Software Engineering

Programming Languages: Python, Java, SQL
Development and Operations: Git, Linux/Shell